

A Quantized Rupture Cosmology: Cyclic Dynamics, Multiverse Branching, and Entropy Inheritance

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Abstract

I propose a rupture-driven cyclic cosmology in which black hole formation increases a singularity density variable $\rho_s(t)$ until a critical threshold ρ_{crit} is reached. Crossing this threshold triggers a rupture that ends the cycle and produces $N = 6m$ descendant universes, where m is a quantized excitation mode determined by rupture strength. Numerical simulations show that for natural parameter choices the fundamental mode $m = 1$ overwhelmingly dominates, yielding six universes per rupture. Extending the model to a branching universe tree and adding entropy tracking reveals a thermodynamic arrow across generations: each universe produces entropy, dumps it into the singularity at rupture, and passes it forward to its descendants. The resulting structure is a stable, self-similar multiverse with quantized reproduction and inherited entropy.

1 Introduction

Cyclic cosmologies have been explored in several contexts, including conformal cyclic cosmology (CCC) [3], ekpyrotic and brane-based cyclic models [4, 5], and loop quantum cosmology (LQC) bounce scenarios [7]. A persistent challenge is identifying a physically motivated mechanism that both terminates each cycle and seeds the next.

In this work, I introduce a rupture-driven cyclic cosmology in which black hole formation drives the accumulation of a singularity density variable $\rho_s(t)$. When $\rho_s(t)$ exceeds a critical threshold ρ_{crit} , the universe undergoes a rupture that resets the cycle and produces a discrete number of descendant universes. A key feature of the model is that rupture outcomes are quantized: the number of universes produced is always a multiple of six, reflecting an assumed six-directional symmetry of the confining structure.

2 Relation to Previous Work

Smolin's cosmological natural selection [1, 2] proposes that black holes generate new universes with varied parameters, forming a branching multiverse. However, Smolin's model is not cyclic, does not include a rupture threshold, and does not impose quantized reproduction.

Penrose's CCC [3] describes an infinite sequence of aeons related by conformal rescaling, with entropy effectively reset between cycles. CCC is cyclic but non-branching: each aeon leads to exactly one successor.

Ekpyrotic models [4, 5] involve brane collisions that periodically reheat the universe, but do not incorporate discrete reproduction or black-hole-driven rupture.

Eternal inflation [6] produces a branching multiverse but lacks cyclic dynamics, a global rupture threshold, or quantized offspring counts.

Loop quantum cosmology [7] replaces singularities with bounces but does not introduce branching or reproduction.

To the author's knowledge, no previous model combines: (i) a singularity density accumulator driven by black hole formation, (ii) a critical rupture threshold, (iii) quantized reproduction in multiples of six, (iv) a branching universe tree, and (v) explicit entropy inheritance across generations.

3 Defining Equation of the Rupture Cosmology

The essential physics of the model can be expressed in a single compact equation:

$$\boxed{N(t) = 6 \Theta(\rho_s(t) - \rho_{\text{crit}}) m(\Delta(t))} \quad (1)$$

where:

- $N(t)$ is the number of universes produced at rupture,
- Θ is the Heaviside step function,
- $\rho_s(t)$ is the singularity density,
- ρ_{crit} is the rupture threshold,
- $\Delta(t) = \rho_s(t) - \rho_{\text{crit}}$ is the overshoot,
- $m(\Delta)$ is a quantized excitation mode.

The excitation mode is defined by discrete overshoot intervals:

$$m(\Delta) = \begin{cases} 1, & \Delta < 500, \text{ } 4pt]2, \\ 500 \leq \Delta < 2000, \text{ } 4pt]3, & 2000 \leq \Delta < 5000, \text{ } 4pt]4, \\ \Delta \geq 5000. \end{cases}$$

The singularity density evolves according to

$$\boxed{\frac{d\rho_s}{dt} = \alpha \dot{M}_{\text{BH}}(t)} \quad (3)$$

with black hole formation rate

$$\boxed{\dot{M}_{\text{BH}}(t) = A t^p e^{-t/\tau}} \quad (4)$$

Equations (1)–(4) define the complete rupture cosmology: black hole formation drives the growth of $\rho_s(t)$ until the threshold ρ_{crit} is crossed, at which point the universe ruptures and produces $N = 6m$ descendant universes determined by the quantized excitation mode m .

4 Universe Tree

Each universe is represented as a node with:

- generation index,
- rupture time,
- entropy produced,
- entropy inherited,
- number of children.

A multiverse tree is built by recursively evolving each universe and spawning $N = 6m$ descendants.

5 Entropy

Entropy production is proportional to black hole formation:

$$S_{\text{produced}} = \int \dot{M}_{\text{BH}}(t) dt. \quad (5)$$

At rupture:

$$S_{\text{dumped}} = S_{\text{produced}} + S_{\text{inherited}}. \quad (6)$$

Each child inherits:

$$S_{\text{inherited}}^{\text{child}} = S_{\text{dumped}}^{\text{parent}}. \quad (7)$$

This produces a thermodynamic arrow across generations.

6 Results

Simulations show:

- stable rupture times ($t_{\text{rupt}} \approx 5.6$ Gyr),
- dominance of the fundamental mode ($m = 1$),
- linear entropy inheritance,
- a stable, self-similar multiverse tree.

7 Conclusion

This rupture-driven cosmology provides a simple mechanism for cyclic evolution, quantized reproduction, and entropy inheritance. The model naturally favors six-universe ruptures, producing a stable branching multiverse with a thermodynamic arrow.

References

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